

Symbol	Any Medium	Lossless Medium ($\epsilon'' = 0$)	Low-loss Medium ($\frac{\epsilon''}{\epsilon'} \ll 1, \frac{\epsilon''}{\epsilon'} < 0.01$)	Good-conductor Medium ($\frac{\epsilon''}{\epsilon'} \gg 1, \frac{\epsilon''}{\epsilon'} > 100$)
$\alpha =$	$\omega \left[\frac{\mu\epsilon'}{2} \left(\sqrt{1 + \left[\frac{\epsilon''}{\epsilon'} \right]^2} - 1 \right) \right]^{1/2}$	0	$\frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\pi f \mu \sigma}$
$\beta =$	$\omega \left[\frac{\mu\epsilon'}{2} \left(\sqrt{1 + \left[\frac{\epsilon''}{\epsilon'} \right]^2} + 1 \right) \right]^{1/2}$	$\omega \sqrt{\mu\epsilon}$	$\omega \sqrt{\mu\epsilon}$	$\sqrt{\pi f \mu \sigma}$
$\eta =$	$\sqrt{\frac{\mu}{\epsilon_c}}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\frac{\mu}{\epsilon}}$	$(1 + j) \frac{\alpha}{\sigma}$
$u_p =$	$\frac{\omega}{\beta}$	$\frac{1}{\sqrt{\mu\epsilon}}$	$\frac{1}{\sqrt{\mu\epsilon}}$	$\sqrt{\frac{4\pi f}{\mu\sigma}}$

$\epsilon' = \epsilon$
$\epsilon'' = \frac{\sigma}{\omega}$
$\epsilon_c = \epsilon' - j\epsilon''$
$u_p = \lambda f$
$\lambda = \frac{2\pi}{\beta}$